

Time: Fri, 5pm–5:50pm, Room 1016

Speaker: Miguel Garrido (ISTA)

Title: Universal Dynamics in a family of Celestial Mechanics models

Abstract: The restricted $(N+1)$ -body problem models the motion of a massless body under the Newtonian gravitational force caused by N other massive bodies, called the primaries. We prove that, for every fixed 2-dimensional symplectic embedding f from the disk to the plane and every fixed accuracy, there exists a restricted planar circular $(N+1)$ -body problem such that a suitable renormalization of its Poincaré map approximates f to that accuracy. The proof relies on Gonchenko-Shilnikov-Turaev universality theory, adapted to a degenerate (parabolic) saddle. This is joint work with Pau Martín (UPC).

Time: Sat, 10am–12pm: space for discussions, common room ground floor (long way straight from the main entrance) and conference room A (next to the common room)

Time: Sat, 12pm–1pm: Lunch, common room ground floor (long way straight from the main entrance)

Time: Sat, 1pm–1:50pm, Room 1016

Speaker: Philipp Gohlke (University of Jena)

Title: Weighted ergodic theorems and spectral theory

Abstract: Going back to work by Wiener–Wintner, weighted ergodic theorems describe the convergence of Birkhoff averages that are modified by a fixed sequence of weights. In this talk we focus on weights that are produced from the sampling of continuous functions along a dynamical system and discuss under which conditions such weights are “good”, that is, the corresponding weighted Birkhoff averages converge. In particular, we will emphasize the role of spectral theory to find sufficient conditions for a dynamically defined sequence to be good. As an application, we show that samplings of primitive, constant length substitution subshifts give rise to good weights. (joint work with M. Neubert and F. Pogorzelski).

Time: Sat, 2pm–2:50pm, Room 1016

Speaker: Tomasz Downarowicz (Wrocław University of Science and Technology)

Title: Induced actions of amenable groups

Abstract: Contrary to the case of a $\mathbb{Z}$ -action if you fix a positive measure set, A in a measure-preserving system (X, μ, G) , where G is a more general amenable group, the returns of points to A typically cannot be organized into orbits of a group action. To overcome this problem, we define the induced action as a skew product over a multiorder - a natural action of the group on the space of linear orders of G of type $\mathbb{Z}$.

Time: Sat, 2:50pm–3:20pm: Coffee break, common room ground floor (long way straight from the main entrance)

Time: Sat, 3:20pm–4:10pm, Room 1016

Speaker: Martha Łącka (Jagiellonian University)

Title: The vague specification property for actions of countable amenable groups

Abstract: I will introduce the vague specification property for actions of countable amenable groups and discuss its relation to weak specification and average shadowing. I will show that weak specification implies vague specification and explain several consequences, including the existence of generic points, completeness of the Besicovitch pseudometric, and the structure of the simplex of invariant measures.

Time: Sat, 4:20pm–5:10pm, Room 1016

Speaker: Alessandra Nardi (ISTA)

Title: Meanders in Rossby Waves

Abstract: Rossby waves, also known as planetary waves, arise naturally in rotating fluids. In the Earth's atmosphere and oceans, they form as a consequence of the planet's rotation. In this setting, meanders are observed in numerical simulations, for instance, transporting cold air masses toward lower latitudes. In this talk, we investigate their possible existence from a mathematical point of view, focusing on the role of the parity of the resonances involved.

Time: Sat, 6:00pm: Dinner at Mucho Gusto

Time: Sun, 9:30am–10:20am, Room 1016

Speaker: Georgios Lamprinakakis (BME Budapest)

Title: Uniform Recurrence for Hyperbolic Automorphisms of $\mathbb{T}^2$

Abstract: We study sets of the form

$$\mathcal{U}(\alpha) := \left\{ x \in X : \exists M = M(x) \geq 1 \text{ such that } \forall N \geq M, \exists 1 \leq n \leq N \text{ with } d(T^n x, x) \leq e^{-\alpha N} \right\},$$

where (X, T, d) is a metric dynamical system. We consider the case $X = \mathbb{T}^2$ and $T(x) = Ax \pmod{1}$, where A is an integer, hyperbolic, area-preserving 2×2 matrix. We explicitly calculate the Hausdorff dimension of this set.

Time: Sun, 10:20am–10:50am: Coffee break, common room ground floor (long way straight from the main entrance)

Time: Sun, 10:50am–11:40am, Room 1016

Speaker: Tobias Jäger (University of Jena)

Title: Structure theory of (multivariate) mean equicontinuous minimal systems

Abstract: We review recent progress in the structure theory of topological dynamics, which has led to a classification of mean equicontinuous minimal systems both in the univariate and multivariate case. It was first shown by Downarowicz and Glasner in 2015 (building on previous work by Li, Tu and Ye) that a minimal topological dynamical system is mean equicontinuous if and only if it is uniquely ergodic and a topo-isomorphic extension of its maximal equicontinuous factor (MEF), meaning that the topological factor map is at the same time a measure-theoretic isomorphism between the two systems. Subsequently, further subclasses defined by dynamical properties (namely frequent stability and diam mean equicontinuity) were equivalently characterised by their extension structures over the MEF (almost one-to-one and regular, respectively). Other dynamical properties like nullness or tameness, can then be related to the resulting hierarchy (they both fall in the regular extension case). Moreover, a topological invariant named amorphic complexity turned out to be tailor-made for the application to mean equicontinuous systems and allows to distinguish conjugacy classes of a large number of examples, including constant length substitutions and inflation tilings.

More recently, this theory has been successfully extended to the multivariate case: a minimal topological dynamical system is mean m -equicontinuous if and only if it is an m -to-one topomorphic extension of its MEF, and similarly frequent m -stability corresponds to an almost m -to-one extension structure and diam mean m -equicontinuity corresponds to m -regularity. The latter is also implied by m -tameness.

The talk aims to give an overview of these recent advances, which are the result of the efforts of a broader community. Specific results by the speaker were obtained in joint work with Felipe Garcia-Ramos and Xiangdong Ye, Eli Glasner and Christian Oertel, Gabriel Fuhrmann, Maik Gröger and Dominik Kwietniak as well as Jonas Breitenbücher and Lino Haupt.
